Kazakh-British Technical University

Data collection and preparation

STUDENTS' INDEPENDENT STUDY (SIS) №1

Analyzing Acoustic Features, Emotional Moods, and Popularity Trends in Music

Full Name: Kabykenova Arailym

ID: 23B030368

Full Name: Alzhan Fariza

ID: 23B031498

09.10.2026

Almaty

1. Introduction

The project studies how acoustic features (tempo, vocal or instrumental) relate to the mood of music and compares this with popularity trends in the Billboard Hot 100 for 2021-2025. Data comes from the Jamendo API and from Wikipedia (web scraping); it was cleaned, merged by year, analyzed and visualized in Python.

2. Data Sources

2.1 API: Jamendo. Endpoint /v3.0/tracks/ with a registered client_id and parameters include=musicinfo, datebetween=2021-01-01_2025-12-31, order=popularity_total, limit=200. With pagination (offset 0-800) we collected 1000 tracks. Because the API returns status 200 even on errors, we checked the status in the headers of each response; empty pages were retried up to 3 times with a pause. Mood tags are sparse, so the mood was taken from the first vartags value instead of filtering the request. Result: jamendo_tracks.csv.

2.2 Web scraping: Wikipedia. Five pages "List of Billboard Hot 100 number ones of 2021-2025" were parsed with requests and BeautifulSoup (User-Agent header, 1 s pause, allowed by robots.txt). Cells merged with rowspan (for example, the same artist for two songs in a row) were copied into every row they cover; each new song cell starts a new No1 run, and its rowspan gives the weeks of that run. Check: the weeks add up to 52 per year (53 in 2022). Result: billboard_hits.csv (113 No1 runs).

3. Data Cleaning and Merging

Step	Jamendo	Billboard
Missing values	2 tracks without mood dropped; missing speed (60), vocal_instrumental (4), genres (2) filled with "unknown"; duration of 0 s (63 tracks) set to missing	none
Duplicates	removed by track_id	exact duplicates removed
Data types	release_date to datetime, year extracted, duration to float (allows missing values)	issue_date + year to datetime
Renaming	track_name, artist to jamendo_track, jamendo_artist	song, artist to billboard_song, billboard_artist
Other	-	footnotes like [note 1], quotes and † marks removed with regex

After cleaning: Jamendo 998 tracks (63 with unknown duration), Billboard 113 No1 runs; no other missing values.

Merging. Jamendo (independent artists) and Billboard (mainstream) share no songs, so we merged by year: Billboard was aggregated to one row per year (number of different No1 songs, number of No1 runs, average and maximum weeks per run) and joined to each track with pd.merge(on="year", how="left"). Result: merged_music.csv (998 rows, 14 columns).

4. Exploratory Data Analysis

Jamendo (998 tracks): average duration 205 s (935 tracks with known duration); tempo is mostly medium (509) and high (293); vocal tracks (552) prevail over instrumental (443). Top moods: love (118), acoustic (64), happy (64), energetic (57), peaceful (51).

Billboard (113 No1 runs): the number of different No1 songs fell from 19 in 2023 to 10 in 2025,

while the average run at No1 grew from 1.86 to 3.47 weeks.

5. Visualizations

Figure 1. Top 6 Track Emotional Moods Distribution

Love dominates (118 tracks); the mood tags are strongly imbalanced toward romantic and positive moods.

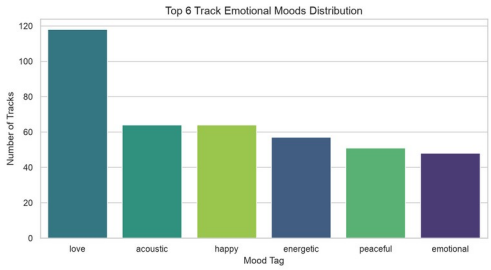


Figure 2. Speed Category Distribution Across Top Track Moods

In almost all moods, tracks are concentrated in medium and high tempo; low and very low tempos are rare.

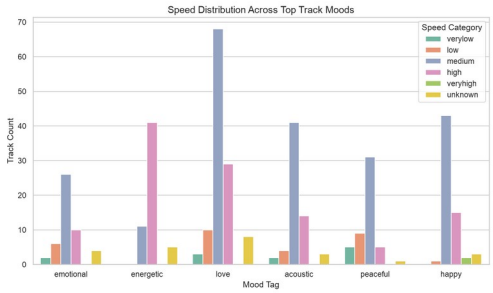
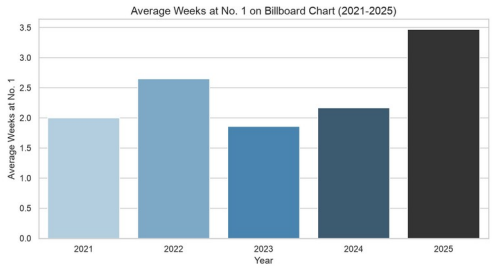


Figure 3. Average Weeks at No. 1 on Billboard Chart (2021-2025)

The average run at No1 grew from 2.00 weeks (2021) to 3.47 weeks (2025): fewer songs reach the top, but they stay longer.



6. Key Insights and Conclusion

- The Billboard chart became less diverse: 2025 had the fewest different No1 songs (10, in 15 runs) and the longest average run (3.47 weeks); the longest runs also grew over time (year vs maximum weeks at No1: r = 0.82).
- Independent tracks prefer medium and high tempo across most moods; only 101 tracks are low tempo.
- The number of No1 songs per year is strongly negatively correlated with average weeks at No1 (r = -0.99, expected since a year has 52 weeks); release year and track duration show only a weak link (r = -0.17).

Conclusion. We built a full pipeline: API collection, web scraping, cleaning, merging, EDA and visualization, and showed how independent music features and mainstream chart dynamics differ. Code and data: https://github.com/arailymkabykenova/SIS1_Music_Analytics